

## Unit 19

# NITROGEN COMPOUNDS

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17 items · 22 marks

### Syllabus sections in this unit

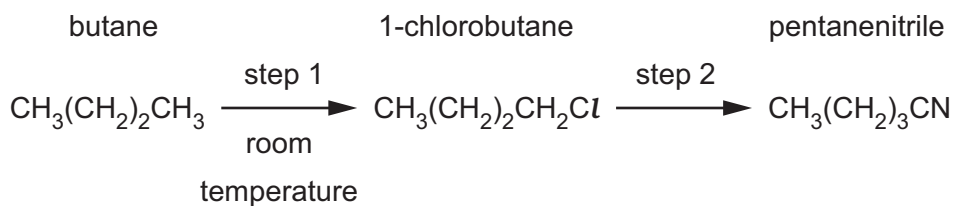
19.1 Primary amines

19.2 Nitriles and hydroxynitriles

**Unit 19: Nitrogen compounds****1. M/J 26/P24/Q4(d)(ii)**

Context: part (d) (introduction)

(d) Butane is converted to pentanenitrile via a two-step synthesis, as shown in Figure 4.3.

**Figure 4.3**

(ii) Pentanenitrile reacts with dilute acid.

Draw the displayed formula of the organic product made in this reaction.

[1]

**2. MAR 26/P22/Q4(a)(iv)**

Context

**4** Cyanoethanoic acid,  $\text{CH}_2(\text{CN})\text{COOH}$ , is a weak organic acid.

Context: part (a) (introduction)

(a) At room temperature,  $\text{CH}_2(\text{CN})\text{COOH}$  undergoes an acid–base reaction with  $\text{NaOH}(\text{aq})$ .

(iv) Suggest the formula of the organic product when  $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$  is heated under reflux with  $\text{NaOH}(\text{aq})$ .

[1]

**3. O/N 25/P24/Q4(b)(ii)**

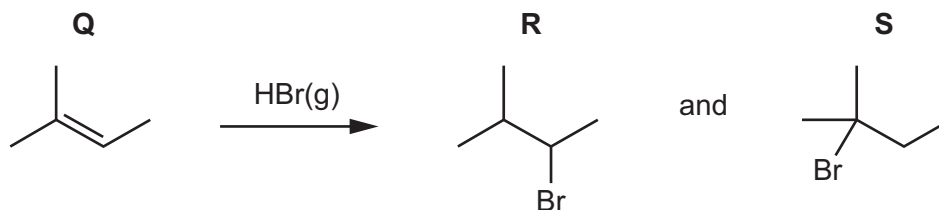
Context

- 4** Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

Context: part (a)(v)

- (v)** **Q** reacts with  $\text{HBr(g)}$  to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

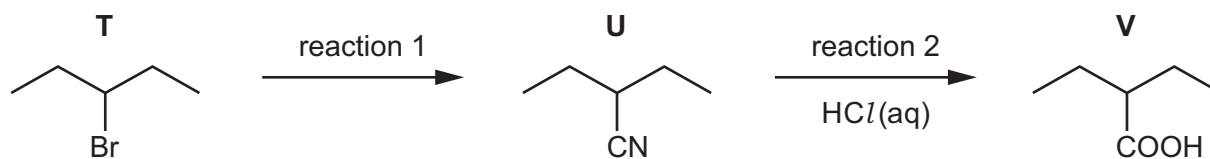
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

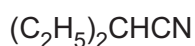
Context: part (b) (introduction)

- (b)** Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

- (ii)** Reaction 2 is a hydrolysis reaction.

Construct an equation for reaction 2.



[1]

**4. M/J 25/P22/Q4(b)(iii)-(iv)**

Context: part (b) (introduction)

- (b)** Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1,  $\text{SOCl}_2$  is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with  $\text{NaOH(aq)}$ .

- (iii)** Construct an equation to describe the reaction of butanenitrile with  $\text{NaOH(aq)}$  in step 3.

[2]

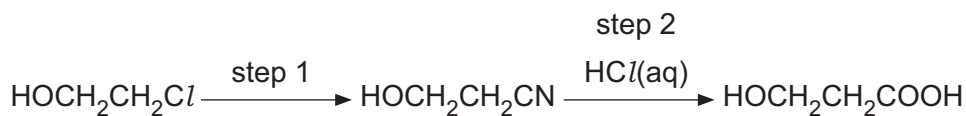
- (iv)** Name the type of reaction that occurs in step 3.

[1]

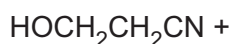
**5. M/J 25/P23/Q6(c)(ii)-(iii)**

Context: part (c) (introduction)

- (c)  $\text{HOCH}_2\text{CH}_2\text{Cl}$  reacts in a two-step synthesis to produce  $\text{HOCH}_2\text{CH}_2\text{COOH}$ , as shown in Fig. 6.2.

**Fig. 6.2**

- (ii) Identify the type of reaction that occurs in step 2.
- (iii) Complete the equation to show the reaction in step 2.



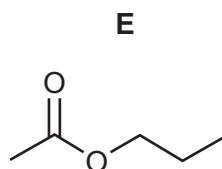
[1]

[1]

**6. M/J 25/P24/Q4(d)(i)**

Context

- 4 The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

Context: part (d) (introduction)

- (d) **F** is an isomer of **E**.

**F** reacts with  $\text{HCl(aq)}$  to produce **G** and **H**.

**G** is a secondary alcohol.

**H** is also produced when  $\text{HCl(aq)}$  is added to  $\text{CH}_3\text{CN}$ .

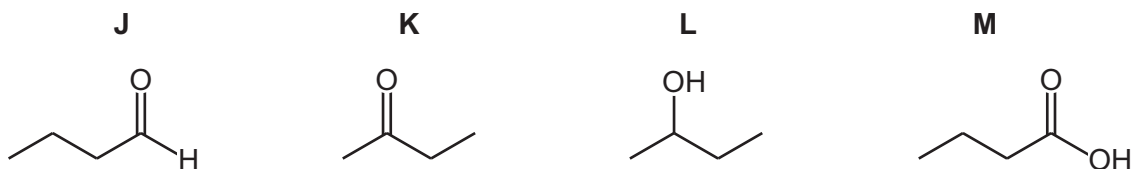
- (i) Construct an equation for the acid hydrolysis of  $\text{CH}_3\text{CN}$  with  $\text{HCl(aq)}$ .

[1]

**7. MAR 25/P22/Q4(b)(ii)**

Context

- 4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

Context: part (b) (introduction)

- (b) **K** reacts with  $\text{HCN}$  in the presence of a  $\text{KCN}$  catalyst, forming **N**.

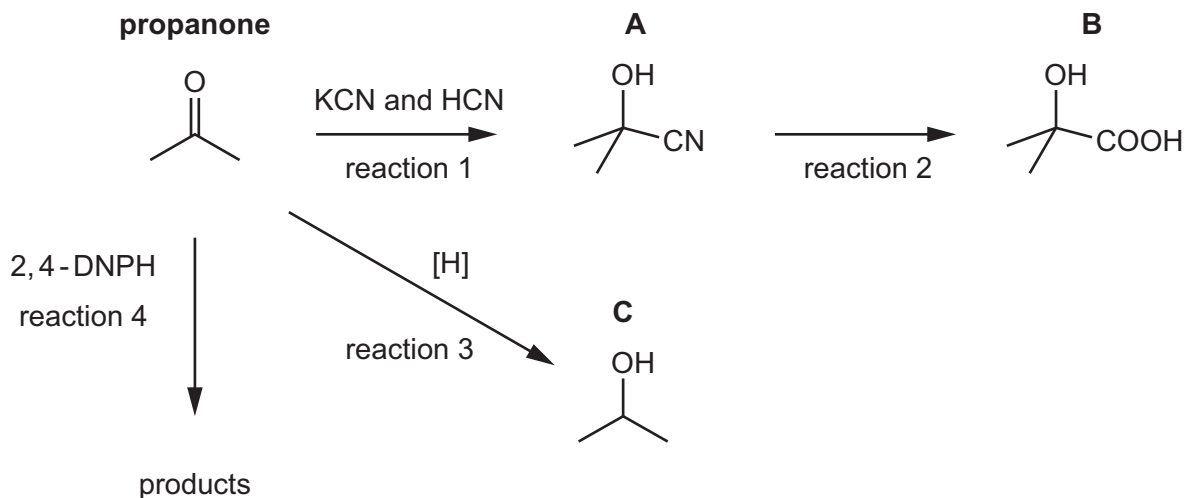
- (ii) Identify the class of compound of which **N** is a member.

[1]

**8. O/N 24/P21/Q4(b)**

Context

- 4** Propanone,  $\text{CH}_3\text{COCH}_3$ , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (b)** Suggest the reagents and conditions for reaction 2.

[1]

**9. O/N 24/P22/Q2(b)(ii)**

Context

- 2** Oxygen is a Group 16 element.

Context: part (b) (introduction)

- (b)**  $\text{H}_2\text{O}$  reacts with both inorganic and organic compounds.

- (ii)** Write an equation for the reaction of  $\text{CH}_3\text{CN}$  with  $\text{H}_2\text{O}$  in acidic conditions.

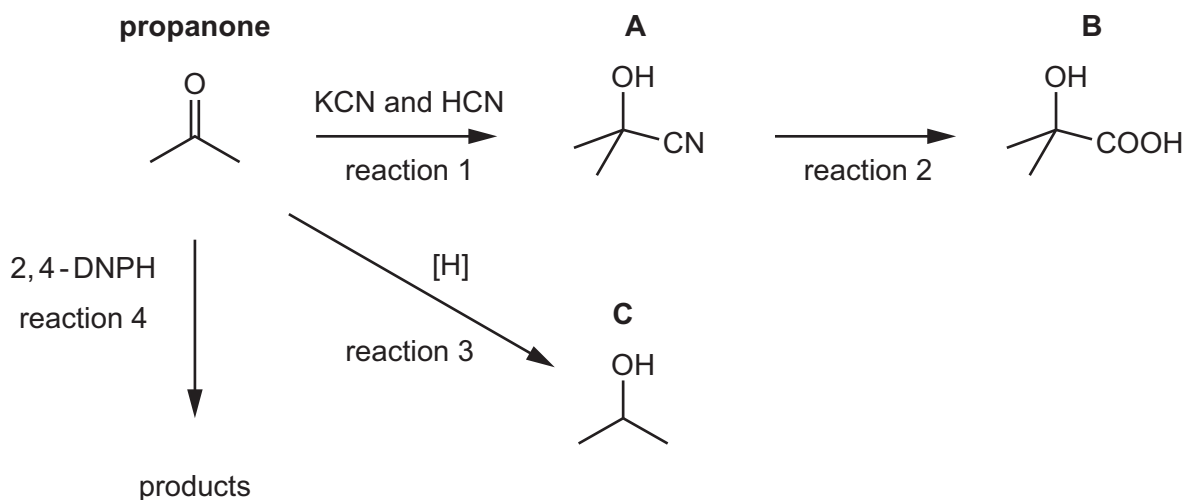


[1]

**10. O/N 24/P23/Q4(b)**

Context

- 4** Propanone,  $\text{CH}_3\text{COCH}_3$ , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

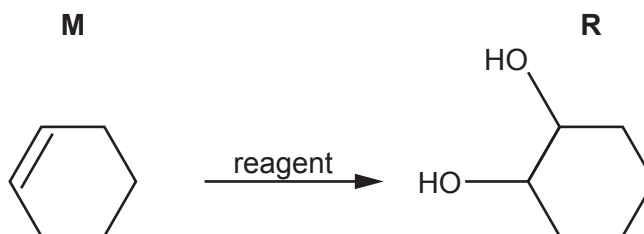
- (b)** Suggest the reagents and conditions for reaction 2.

[1]

**11. M/J 24/P23/Q5(d)(i)-(ii)**

Context: part (a)

- 5** **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

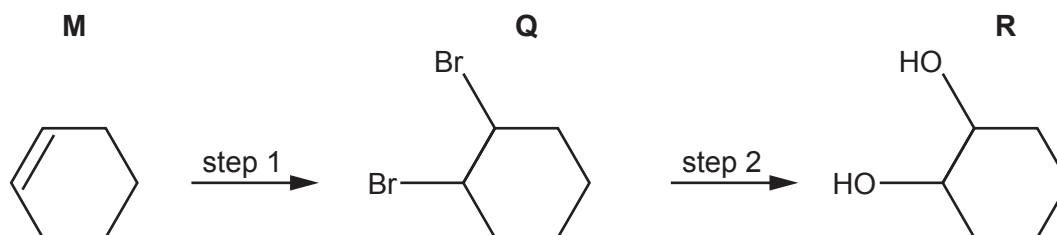
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

Context: part (d) (introduction)

- (d)** **Y** is made from **Q** in a three-step reaction.

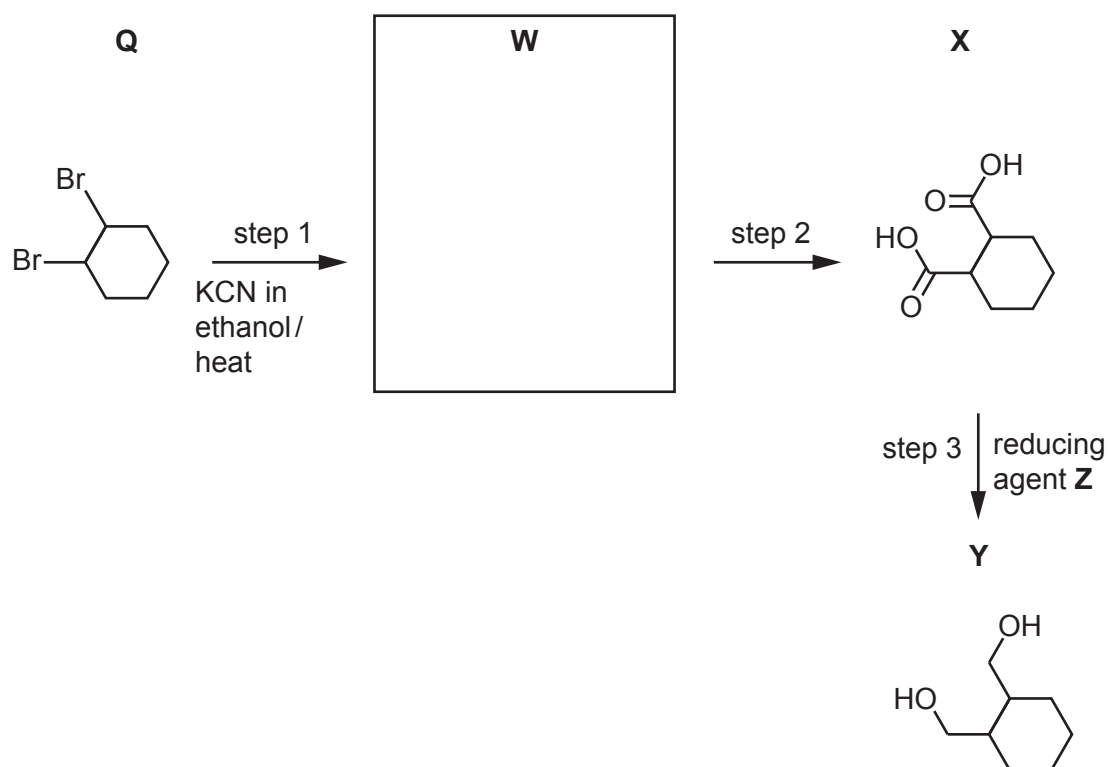


Fig. 5.4

- (i) Draw the structure of **W** in the box in Fig. 5.4. [1]
- (ii) In step 2, **W** is heated with  $\text{HCl(aq)}$  to produce **X** and an inorganic product. Identify the formula of the inorganic product. [1]

## 12. MAR 24/P22/Q4(b)(iii)

Context

- 4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

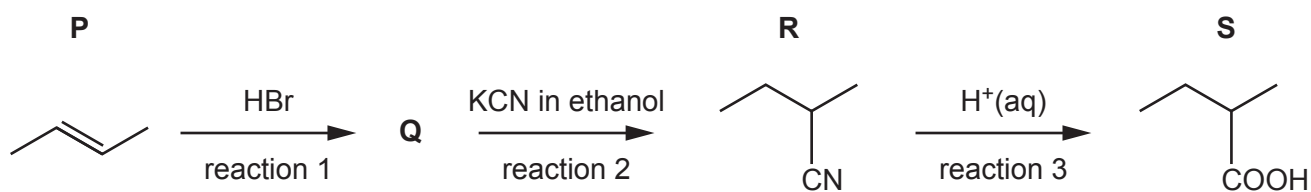
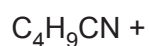


Fig. 4.1

- (iii) Complete the equation for reaction 3. **R** is represented as  $\text{C}_4\text{H}_9\text{CN}$ .

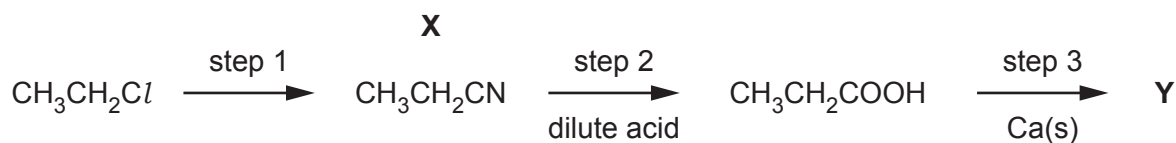


[1]

**13. M/J 23/P22/Q3(a)(iii)**

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from  $\text{CH}_3\text{CH}_2\text{Cl}$ .

**Fig. 3.1**

- (iii) Identify the type of reaction that occurs when dilute acid is added to **X** in step 2.

[1]

**14. M/J 23/P23/Q6(e)**

Context

- 6** Compound **W**,  $\text{CH}_2=\text{CHCN}$ , is used to make an addition polymer which is present in carbon fibres.

- (e) Propanenitrile is heated with hydrogen gas and a platinum catalyst. The only product is propylamine.

Construct an equation for this reaction.

[1]

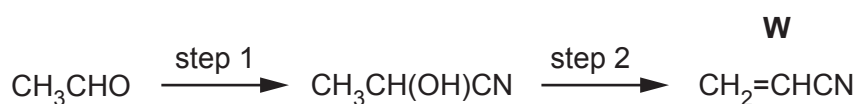
**15. M/J 23/P23/Q6(f)(ii)**

Context

- 6** Compound **W**,  $\text{CH}_2=\text{CHCN}$ , is used to make an addition polymer which is present in carbon fibres.

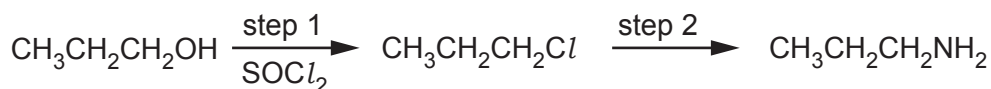
Context: part (b) (introduction)

- (b)  $\text{CH}_3\text{CHO}$  is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1. In step 1,  $\text{CH}_3\text{CHO}$  is heated with  $\text{HCN}$  in the presence of  $\text{KCN}$ .

**Fig. 6.1**

Context: part (f) (introduction)

- (f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

**Fig. 6.4**

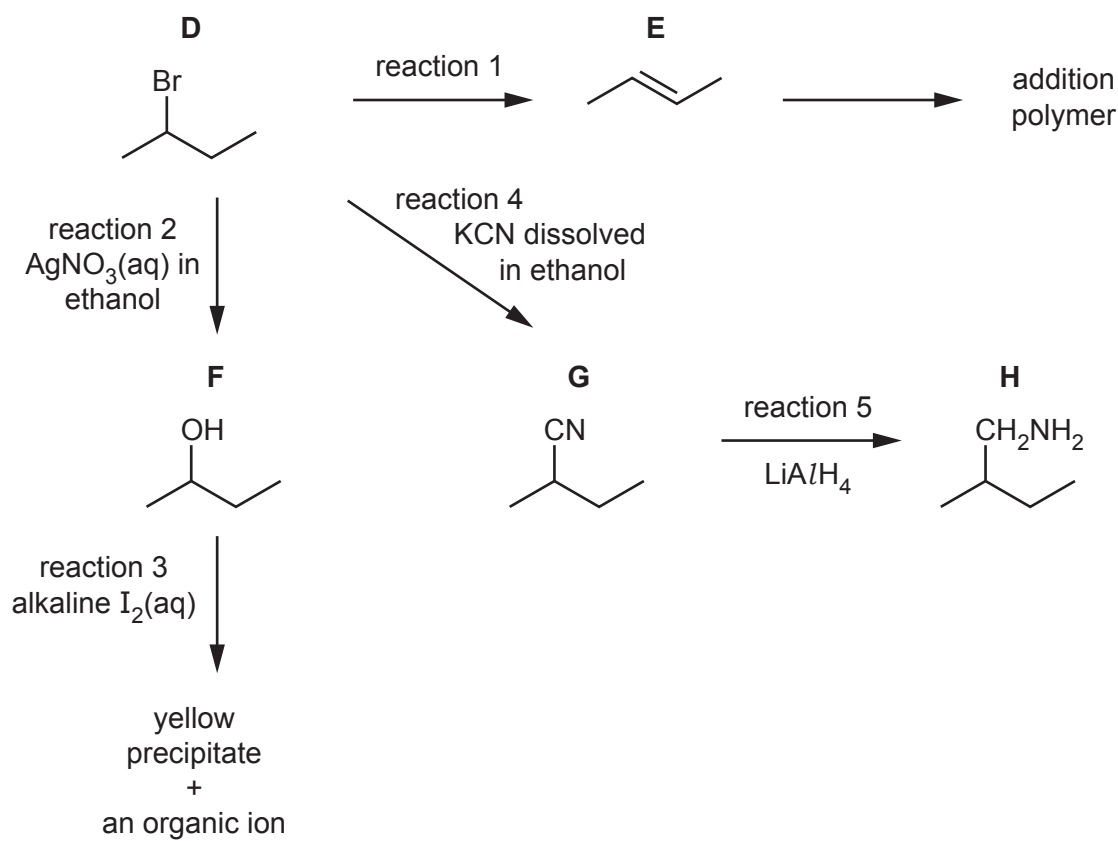
- (ii) Identify the reagent and conditions for step 2 in Fig. 6.4.

[2]

**16. MAR 23/P22/Q4(c)(ii)**

Context

**4** Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.



**Fig. 4.1**

(ii) Reaction 5 is similar to the reaction of  $\text{LiAlH}_4$  with carboxylic acids to form alcohols.

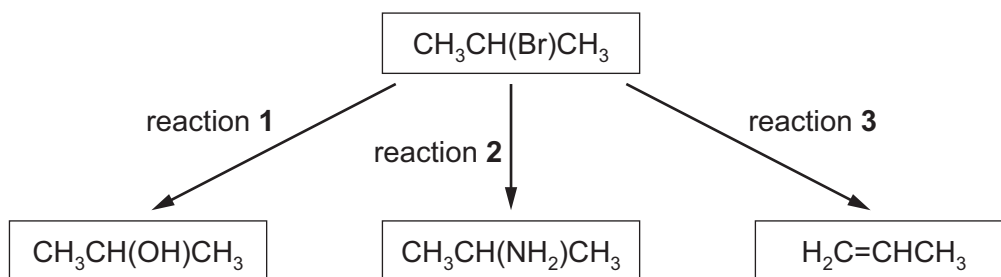
Suggest the role of  $\text{LiAlH}_4$  in reaction 5.

[1]

**17. M/J 22/P22/Q5(c)(ii)**

Context

**5** Fig. 5.1 shows three reactions of 2-bromopropane,  $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$ .

**Fig. 5.1**

Context: part (c) (introduction)

**(c)** Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.

**Fig. 5.2**

**(ii)** In step 2, butanenitrile is heated with  $\text{HCl}(\text{aq})$ . A hydrolysis reaction occurs.

Construct an equation for the reaction in step 2.

[1]

## Answers Section

### 1. M/J 26/P24/Q4(d)(ii)

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
| 4(d)(ii) | $  \begin{array}{ccccccc}  & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \\  &   & &   & &   & &   & & \\  \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\  &   & &   & &   & &   & & \\  & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \\  & & & & & & & & & \text{O} \\  & & & & & & & & & // \\  & & & & & & & & & \text{O} \\  & & & & & & & & & \backslash \\  & & & & & & & & & \text{O} - \text{H}  \end{array}  $ | 1 | R-OH |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|

### 2. MAR 26/P22/Q4(a)(iv)

|          |                                                                                                   |   |
|----------|---------------------------------------------------------------------------------------------------|---|
| 4(a)(iv) | CH <sub>2</sub> (COO) <sub>2</sub> Na <sub>2</sub> <b>OR</b> CH <sub>2</sub> (COONa) <sub>2</sub> | 1 |
|----------|---------------------------------------------------------------------------------------------------|---|

### 3. O/N 25/P24/Q4(b)(ii)

|          |                                                                                                                                                          |   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(b)(ii) | (C <sub>2</sub> H <sub>5</sub> ) <sub>2</sub> CHCN + 2H <sub>2</sub> O + HCl → (C <sub>2</sub> H <sub>5</sub> ) <sub>2</sub> CHCOOH + NH <sub>4</sub> Cl | 1 |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Answer for context part (a)(v)

|         |                                                                                                                                                           |   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(a)(v) | <b>M1</b> (reaction proceeds via a) more stable intermediate / carbocation<br><b>M2</b> because of greater positive inductive effect of more alkyl groups | 2 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|

### 4. M/J 25/P22/Q4(b)(iii)-(iv)

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(b)(iii) | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CN + NaOH + H <sub>2</sub> O → CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CO <sub>2</sub> Na + NH <sub>3</sub><br><b>OR</b><br>CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CN + NaOH + 2H <sub>2</sub> O → CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CO <sub>2</sub> Na + NH <sub>4</sub> OH<br><b>M1</b> correct organic product (as molecule or anion)<br><b>M2</b> rest of equation is correct | 2 |
| 4(b)(iv)  | (alkaline) hydrolysis                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1 |

### 5. M/J 25/P23/Q6(c)(ii)-(iii)

|           |                                                                                                                              |   |
|-----------|------------------------------------------------------------------------------------------------------------------------------|---|
| 6(c)(ii)  | (acid) hydrolysis                                                                                                            | 1 |
| 6(c)(iii) | HOCH <sub>2</sub> CH <sub>2</sub> CN + 2H <sub>2</sub> O + HCl → HOCH <sub>2</sub> CH <sub>2</sub> COOH + NH <sub>4</sub> Cl | 1 |

### 6. M/J 25/P24/Q4(d)(i)

|         |                                                                                          |   |
|---------|------------------------------------------------------------------------------------------|---|
| 4(d)(i) | CH <sub>3</sub> CN + HCl + 2H <sub>2</sub> O → CH <sub>3</sub> COOH + NH <sub>4</sub> Cl | 1 |
|---------|------------------------------------------------------------------------------------------|---|

### 7. MAR 25/P22/Q4(b)(ii)

|          |                |   |
|----------|----------------|---|
| 4(b)(ii) | hydroxynitrile | 1 |
|----------|----------------|---|

### 8. O/N 24/P21/Q4(b)

|      |                                                |   |
|------|------------------------------------------------|---|
| 4(b) | H <sub>2</sub> SO <sub>4</sub> (aq) or HCl(aq) | 1 |
|------|------------------------------------------------|---|

### 9. O/N 24/P22/Q2(b)(ii)

|          |                                                                                                                  |   |
|----------|------------------------------------------------------------------------------------------------------------------|---|
| 2(b)(ii) | CH <sub>3</sub> CN + 2H <sub>2</sub> O + (1)H <sup>+</sup> → CH <sub>3</sub> COOH + NH <sub>4</sub> <sup>+</sup> | 1 |
|----------|------------------------------------------------------------------------------------------------------------------|---|

### 10. O/N 24/P23/Q4(b)

|      |                                                |   |
|------|------------------------------------------------|---|
| 4(b) | H <sub>2</sub> SO <sub>4</sub> (aq) or HCl(aq) | 1 |
|------|------------------------------------------------|---|

**11. M/J 24/P23/Q5(d)(i)-(ii)**

|          |                        |          |
|----------|------------------------|----------|
| 5(d)(i)  | <b>W</b> =             | <b>1</b> |
| 5(d)(ii) | $\text{NH}_4\text{Cl}$ | <b>1</b> |

Answer for context part (a)

|      |                                                                            |          |
|------|----------------------------------------------------------------------------|----------|
| 5(a) | cold dilute $\text{KMnO}_4$ <b>OR</b> cold dilute potassium manganate(VII) | <b>1</b> |
|------|----------------------------------------------------------------------------|----------|

**12. MAR 24/P22/Q4(b)(iii)**

|           |                                                                                                                                |          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(b)(iii) | $\text{C}_4\text{H}_9\text{CN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{C}_4\text{H}_9\text{COOH} + \text{NH}_4^+$ | <b>1</b> |
|-----------|--------------------------------------------------------------------------------------------------------------------------------|----------|

**13. M/J 23/P22/Q3(a)(iii)**

|           |                   |          |
|-----------|-------------------|----------|
| 3(a)(iii) | (acid) hydrolysis | <b>1</b> |
|-----------|-------------------|----------|

**14. M/J 23/P23/Q6(e)**

|      |                                                                                                          |          |
|------|----------------------------------------------------------------------------------------------------------|----------|
| 6(e) | $\text{CH}_3\text{CH}_2\text{CN} + 2\text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ | <b>1</b> |
|------|----------------------------------------------------------------------------------------------------------|----------|

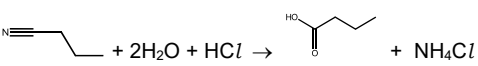
**15. M/J 23/P23/Q6(f)(ii)**

|          |                                                                                      |          |
|----------|--------------------------------------------------------------------------------------|----------|
| 6(f)(ii) | <b>M1</b> ammonia / $\text{NH}_3$<br><b>M2</b> heat + (M1 in ethanol) under pressure | <b>2</b> |
|----------|--------------------------------------------------------------------------------------|----------|

**16. MAR 23/P22/Q4(c)(ii)**

|          |                |          |
|----------|----------------|----------|
| 4(c)(ii) | reducing agent | <b>1</b> |
|----------|----------------|----------|

**17. M/J 22/P22/Q5(c)(ii)**

|          |                                                                                                                                                                                                                                                       |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(c)(ii) | $\text{CN}(\text{CH}_2)_2\text{CH}_3 + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{HOOC}(\text{CH}_2)_2\text{CH}_3 + \text{NH}_4\text{Cl}$<br><b>OR</b><br> | <b>1</b> |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|